

Xinyi Zhu

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EDUCATION

Carnegie Mellon University

Master of Science in Information Networking

Pittsburgh, PA

Expected Dec 2027

University of Michigan-Shanghai Jiao Tong University Joint Institute

Bachelor of Engineering in Mechanical Engineering

Shanghai, China

Sep 2022 - Aug 2026

University of Michigan

Bachelor of Science in Computer Science

Ann Arbor, MI

Aug 2024 - May 2026

GPA 4.00/4.00 | Coursework: Web Systems (A+), Operating Systems (A+), System Design of a Search Engine (A+), Machine Learning (A+), Natural Language Processing (A+), Reinforcement Learning (A)

SKILLS

Programming Languages: Python, C++, SQL, Go, JavaScript, Bash, C, MATLAB

AI/ML & Agents: PyTorch, LangGraph, LangChain, DSPy, GEPA, NumPy, pandas, Weights & Biases, FAISS

Software & Infrastructure: FastAPI, Node.js, React, REST APIs, HTML/CSS, Git, Linux, Docker, AWS, pytest, Neo4j

WORK EXPERIENCE

ByteDance

Shanghai, China

AI Agent Backend Engineer Intern

May 2026 - Aug 2026

- Designed and launched an evaluation framework for an LLM-powered on-call assistant, combining user-adoption signals with LLM-as-a-judge scoring; evaluated 2,900+ response rounds across 1,300+ support chats and surfaced low-quality cases for targeted iteration.
- Designed and implemented a Docker Buildx/BuildKit-based WebShell debugging system that decoupled build execution from user sessions, preserving failed-build root filesystems in a shared debug session while providing each engineer or AI agent an isolated tmux shell for collaborative troubleshooting.
- Extended a multi-agent on-call troubleshooting system for container-image build failures by developing scenario-specific SOP-as-code workflows, refining agent routing and prompts, and integrating build-context tools; increased the AI contribution score by 15.3% and reduced MTTR by 11.4%.
- Built a self-evolving LLM knowledge base that distilled 10K+ support chats into confidence-scored Q&A by clustering questions and consistent answers, while using human feedback to resolve conflicts and refresh stale knowledge; recovered 40.1% of unanswered queries and raised useful-response rate from 69.3% to 73.9%.

PROJECT EXPERIENCE

Distributed Search Engine | C++, Distributed Systems, Information Retrieval

Jan 2026 - May 2026

- Built a high-performance C++ indexing pipeline with compressed inverted indexes using delta and variable-byte encoding, reducing index storage by 40%.
- Distributed indexing and retrieval across 10 nodes using index sharding, parallel local ranking, and scatter-gather aggregation, scaling to 100M documents at over 1M indexed documents per second.
- Ranked 1st among 14 teams in the capstone showcase for search relevance and retrieval efficiency.

Polaris: Graph-Based Research Index & Retrieval System | Python, Neo4j, Cypher, FAISS

May 2025 - Mar 2026

- Built a Neo4j-based knowledge graph and data ingestion pipeline representing concepts, methods, and relationships across 60K+ computer science papers.
- Developed a tool-using retrieval agent that combined Cypher graph queries with vector search and employed beam-search-guided multi-path reasoning to identify relevant scientific literature.

Advanced Operating Systems | C++, Boost, TCP/IP

Jan 2025 - May 2025

- Implemented a single-CPU user-level thread library using ucontext, with FIFO scheduling, timer-interrupt preemption, thread joins, mutexes, and condition variables.
- Built a demand-paged virtual memory manager supporting swap-backed and file-backed pages, clock-based eviction, dirty/reference tracking, eager swap reservation, and copy-on-write from a pinned zero page.
- Built a multithreaded TCP RPC file server supporting hierarchical path traversal and concurrent file creation, deletion, reading, and writing, with inode-level shared/exclusive locking and synchronized free-block allocation.

EXP-Bench: Can AI Conduct AI Research Experiments? (ICLR 2026) | LLMs, Data Pipelines

Mar 2025 - May 2025

- Developed a semi-automated pipeline to extract experimental procedures, configurations, and evaluation criteria from research papers, supplementary materials, and code repositories.
- Contributed to a scalable benchmark that reconstructs executable research tasks from scientific literature to evaluate AI agents on real-world experimentation.